

HERNAN “ERNIE” MALDONADO

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Summary

ML engineer and data scientist with 10+ years delivering production systems in fraud detection, sanctions compliance, and cybersecurity at major financial institutions. Specialist in NLP pipelines, gradient-boosted models, and Hadoop-scale data engineering. Holder of three graduate degrees (computational mathematics, public policy, economics) and four ML/deep-learning certifications. Targeting ML, generative AI, or data engineering roles.

Work Experience

Citigroup, Tampa FL

Sep 2024 – Present

Internal Fraud Engineer

Hybrid

- Mine Hadoop databases (transactions, user behavior, historical cases) to surface fraud patterns and translate findings into PySpark/Hive detection rules targeting anomalous velocity, high-risk access, and suspicious vendors.
- Collaborate with analysis teams to simulate, tune, and deploy alert logic; manage alert volume by focusing on recurring MOs that mirror confirmed past crimes.

Bank of America, Tampa FL

Oct 2023 – Dec 2023

Consumer Protection Big Data & ML Consultant

Hybrid

- Audited the entire fraud Hadoop ecosystem and produced a data-driven benchmark report comparing current practices against industry standards across performance, reliability, and data quality.

Citigroup, Tampa FL

May 2022 – Oct 2023

Sanctions Machine Learning Engineer

Hybrid

- **Automated 500K monthly sanctions alerts** by building end-to-end NLP pipelines (Random Forest, Gradient Boosting, XGBoost) with 50+ engineered features to match flagged customers against sanctioned-persons lists.
- **Cut manual review volume by 50%** by discovering Citi's internal transactions were being double-screened across two independent systems.
- Authored Model Risk Management documentation ensuring compliance with OCC 2011-12 and SR 11-7; produced architecture study showing a Hadoop migration would reduce processing time by **90%**.
- *Stack: SQL, Pandas, Python, Scikit-Learn, PyTorch, XGBoost, FastAPI*

Certegy Payments, Tampa FL

May 2021 – Apr 2022

Data Scientist – Fraud Detection ML

Hybrid

- **Doubled check-fraud detection rate** at Walmart by developing Random Forest, Gradient Boosting, and Logistic Regression models; raised KS statistic from 0.05–0.3 to 0.6–0.8.
- Architected full ML pipeline: features built in Oracle SQL, models served via AWS API with real-time and batch scoring, routed to the correct model per transaction type.
- *Stack: SAS Viya, Oracle SQL, Oracle OML, Python, PyTorch, Scikit-Learn, TensorFlow, AWS*

Nielsen, Tampa FL

Feb 2021 – Apr 2021

Sr. Data Scientist

Remote

- Maintained and extended Java codebase for national-scale TV viewership adjudication; developed new attribution features for programming and advertising data. *Stack: Java, S3, Git*

Citigroup, Tampa FL

Apr 2018 – Feb 2021

Cybersecurity Big Data & ML Engineer

Hybrid

- **Built from scratch** a production anomaly-detection system monitoring **300K employees** across **5TB of weekly data** using peer-normalized behavioral metrics and k-means clustering to surface insider threats for the Cyber Security Fusion Center.
- Engineered ingestion pipelines for **12+ data sources** into a **270TB Hadoop cluster** (Sqoop, Flume, Python parsers, Hive, Oozie) and built a monitoring system to ensure all feed flows remained current.
- *Stack: PySpark, PyTorch, Scikit-Learn, TensorFlow, Hive, Oozie, Hadoop, Python, SQL, Linux*

Wellcare Health Plans (Centene), Tampa FL

May 2016 – Oct 2017

Data Scientist – Machine Learning

On-Site

- **Reduced hospital readmissions by 60%** using XGBoost and Scikit-Learn, directly limiting Medicare penalty exposure for the plan.
- Key contributor to a **\$1M/year capital project** coordinating Informatica, Hadoop, SAS, and Python workflows across external vendors and multiple business units.
- *Stack: PySpark, Hive, Python, R, SQL, XGBoost, Scikit-Learn*

Gateway HealthPlan (Highmark), Pittsburgh PA

Jan 2015 – Apr 2016

Sr. Decision Support Analyst

On-Site

- Worked directly with the CFO to resolve reconciliation gaps between insurance claims and accounting systems via ETL improvements; built ARIMA cash-flow forecasting model in R; managed a team of three data engineers and supervised SAS platform administration.

UPMC Health Plan, Pittsburgh PA

Aug 2012 – Oct 2014

Database & ETL Analyst

On-Site

- Built enterprise-wide ETL reporting layers in Oracle SQL and SAS; developed Random Forest model to segment high-cost members. Earned the **2014 ACES Award** (top 1% of UPMC employees) for excellence in service and process improvement.

University of Pittsburgh, Pittsburgh PA

Jan 2011 – Aug 2012

Project Manager

On-Site

- Managed two \$1.5M grant-funded research projects end-to-end (20+ staff), completed data collection ahead of schedule and under budget; projects featured on local television.

Projects

NameGen: Generative AI Name Generator <i>Python, FastAPI, Docker</i>	Source Code
- Generates novel names using directed-graph letter-transition probabilities sampled via multinomial distribution — the same statistical foundation underlying language model sampling. - Deployed three ways: interactive CLI (with pickle-cached graph for speed), local FastAPI service, and Dockerized container, demonstrating a complete ML deployment lifecycle.	
E-Commerce Fraud Detection <i>Python, XGBoost, Optuna, joblib</i>	Source Code
Three-model ensemble (Random Forest, Isolation Forest, XGBoost) applied to e-commerce fraud; XGBoost hyperparameters tuned with Optuna. Models serialized with joblib pipelines ready for production deployment.	

Education

Duquesne University, Pittsburgh PA USA	2008 - 2011
<i>Masters in Computational Mathematics</i>	<i>3.9 GPA</i>
KDI School, Seoul (Relocated to Sejong City), S. Korea	2004 - 2005
<i>Masters in Public Policy</i>	<i>3.9 GPA</i>
Universidad de los Andes, Bogota, Colombia	2001 - 2003
<i>Masters in Economics</i>	<i>3.7 GPA</i>
Universidad de los Andes, Bogota, Colombia	1996 - 2001
<i>Bachelors in Economics</i>	<i>3.7 GPA</i>

Certificates

AWS Certified Cloud Practitioner	2022
Tensor Flow Specialization	2020
Deep Learning Specialization	2018
Machine Learning Specialization	2017
Machine Learning Certificate	2015

Technical Skills

Languages: Python, Java, SAS, SQL, R, Rust, Matlab
Machine Learning: Scikit-Learn, PyTorch, TensorFlow, XGBoost, Optuna, SAS Viya, PySpark MLlib
GenAI & NLP: LLM APIs, prompt engineering, NLP feature engineering, generative modeling
Big Data: Apache Hadoop, Hive, PySpark, Oozie, Impala, Sqoop, Flume
Cloud & Databases: AWS, Oracle DB, MS SQL Server, MySQL
MLOps & Tools: FastAPI, Docker, Flask, Git, GitHub, GitLab, BitBucket, Postman, Jira

Selected Publications

Participatory Assessment of the Health of Latino Immigrant Men in a Community	2013
<i>Journal of Immigrant and Minority Health</i>	
Bayesian Regression Inference Using a Normal Mixture Model	2012
<i>Masters Thesis, Duquesne University</i>	
FARC Terrorism in Colombia: A Clustering Analysis	2004
<i>CEDE Working Papers Series</i>	